Shocks, Weak Solutions, and Entropy: An Exposition of Burgers' Equation

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1 Introduction

The viscous Burgers' equation

$$u_t + u\,u_x = \nu\,u_{xx}, \qquad \nu > 0, \tag{1}$$

was introduced by Burgers [2] as a one-dimensional model of turbulence; it had appeared earlier in work of Bateman [1]. It has the form of the Navier–Stokes momentum equation without the pressure term, and combines nonlinear advection with viscous diffusion. Setting $\nu = 0$ gives the inviscid equation

$$u_t + u\,u_x = 0, \tag{2}$$

the scalar conservation law $u_t + f(u)_x = 0$ with flux $f(u) = u^2/2$. Conservation laws of this type arise in gas dynamics [5, 10] and traffic flow [11]. Burgers' equation is the simplest model in which nonlinearity alone produces discontinuities: it is the standard test case for shock-capturing numerical methods, and it is one of the few nonlinear equations that can be solved exactly, through the Cole–Hopf transformation, so that every claim about shocks and entropy can be checked against an explicit formula.

Classical solutions of (2) can break down in finite time for smooth data (as soon as u_0' is negative somewhere; see Section 3), beyond which they can be continued only in the weak sense, and weak solutions are not unique. Since (2) idealises (1) for small ν, with a shock appearing as a layer of thickness $O(\nu)$, the admissible weak solutions should be those that arise as limits of viscous solutions. We therefore ask *why the inviscid Burgers' equation requires an entropy condition, and why that condition is the correct one.* The answer is that the entropy condition is the criterion selected by vanishing viscosity, a result due to Hopf (Section 8). Sections 2 and 3 locate the breakdown of classical solutions, and Sections 4–7 develop weak solutions, their non-uniqueness and the entropy conditions that restore uniqueness, culminating in the Riemann problem; Section 8 connects these with the Cole–Hopf formula, vanishing viscosity and Godunov's method. The theory follows [5, §3.4].

2 Method of Characteristics

Consider the initial value problem

$$\begin{cases} u_t + u\,u_x = 0, & x \in \mathbb{R},\ t > 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \tag{3}$$

with $u_0 \in C^1(\mathbb{R})$ in Sections 2 and 3. For discontinuous data, such as the Riemann data (19), (3) is understood in the weak sense of Definition 4.1.

2.1 Characteristics

A *characteristic* of a C^1 solution u is a curve $x(t)$ satisfying

$$\dot{x}(t) = u(x(t), t). \tag{4}$$

Along it,

$$\frac{d}{dt}\,u(x(t),t) = u_x\,\dot{x} + u_t = u_t + u\,u_x = 0. \tag{5}$$

Consequently u is constant along each characteristic, so $\dot{x}$ is constant: if u is a C^1 solution of (3) on $\mathbb{R}\times[0,T)$ and $x(t)$ solves (4) with $x(0)=x_0$, then for $0\le t<T$

$$u(x(t),t)=u_0(x_0),\qquad x(t)=x_0+u_0(x_0)\,t, \tag{6}$$

that is, characteristics are straight lines.

2.2 The implicit solution formula

Theorem 2.1 (Implicit solution via characteristics)**.** *Let $u_0\in C^1(\mathbb{R})$, $\Phi_t(x_0)=x_0+u_0(x_0)\,t$, and $T\in(0,\infty]$ such that for every $t\in[0,T)$, $\Phi_t'>0$ on $\mathbb{R}$ and Φ_t maps $\mathbb{R}$ onto $\mathbb{R}$. Then the function u defined implicitly by*

$$u(x,t)=u_0(x_0),\qquad \text{where } x=x_0+u_0(x_0)\,t, \tag{7}$$

is the unique C^1 solution of (3) on $\mathbb{R}\times[0,T)$.

Proof. Well-definedness and regularity. Since $\Phi_t'>0$, Φ_t is strictly increasing, hence bijective by hypothesis, so $u(x,t)=u_0(\Phi_t^{-1}(x))$ is well defined. The function $F(x_0,x,t)=x_0+u_0(x_0)t-x$ is C^1 with $\partial_{x_0}F=\Phi_t'>0$, so by the implicit function theorem $x_0(x,t)=\Phi_t^{-1}(x)$ is C^1 in (x,t) jointly (it is locally the unique root, and by injectivity coincides with the global one). Hence $u=u_0(x_0(x,t))$ is C^1.

The PDE holds. Write $x_0=x_0(x,t)=\Phi_t^{-1}(x)$. Differentiating $x=x_0+u_0(x_0)\,t$ in x and in t gives

$$\frac{\partial x_0}{\partial x}=\frac{1}{1+u_0'(x_0)\,t},\qquad \frac{\partial x_0}{\partial t}=\frac{-u_0(x_0)}{1+u_0'(x_0)\,t}, \tag{8}$$

both well defined since $\Phi_t'(x_0)=1+u_0'(x_0)t>0$. By the chain rule applied to $u(x,t)=u_0(x_0(x,t))$,

$$u_x=\frac{u_0'(x_0)}{1+u_0'(x_0)\,t},\qquad u_t=\frac{-u_0(x_0)\,u_0'(x_0)}{1+u_0'(x_0)\,t}, \tag{9}$$

so $u_t+u\,u_x=0$, and $u(x,0)=u_0(x)$ since $\Phi_0=\mathrm{id}$.

Uniqueness. Let v be any C^1 solution of (3) on $\mathbb{R}\times[0,T)$, and fix $t\in(0,T)$, $x\in\mathbb{R}$. Solve backwards in time the ODE $\dot\gamma(s)=v(\gamma(s),s)$, $\gamma(t)=x$ (the right-hand side is C^1, hence locally Lipschitz, so there is a unique maximal solution). By (5), v is constant along γ, hence so is $\dot\gamma$: γ is a straight line with bounded slope $v(x,t)$. It therefore stays bounded and cannot blow up, so it extends back to $s=0$. Setting $x_0=\gamma(0)$ we get $v(x,t)=v(x_0,0)=u_0(x_0)$ and $x=x_0+u_0(x_0)t$. Thus every point (x,t) lies on a characteristic coming from $t=0$, with $x_0=\Phi_t^{-1}(x)$, and $v(x,t)=u_0(\Phi_t^{-1}(x))=u(x,t)$. □

Corollary 2.2 (Local existence)**.** *Let $u_0\in C^1(\mathbb{R})$ with $m:=\inf_{x_0}u_0'(x_0)>-\infty$, and set $T=\infty$ if $m\ge0$ and $T=-1/m$ if $m<0$. Then (3) has a unique C^1 solution on $\mathbb{R}\times[0,T)$, given by (7).*

Proof. Fix $t\in[0,T)$ and put $\kappa_t:=1+\min(m,0)\,t$, so $\kappa_t>0$ and $\Phi_t'(x_0)=1+u_0'(x_0)t\ge\kappa_t$ for every x_0. Two consequences follow. (i) Φ_t is strictly increasing, hence injective. (ii) By the mean value theorem $\Phi_t(x_0)-\Phi_t(0)\ge\kappa_t x_0$ for $x_0\ge0$ and $\le\kappa_t x_0$ for $x_0\le0$, so $\Phi_t(x_0)\to\pm\infty$ as $x_0\to\pm\infty$; by continuity Φ_t is onto $\mathbb{R}$. Hence the hypotheses of Theorem 2.1 hold on $[0,T)$. □

Remark 2.3. The uniform lower bound $\Phi_t'\ge\kappa_t>0$, not strict monotonicity alone, gives surjectivity: for $u_0(x)=(\arctan x-x)/t_0$ one has $\Phi_{t_0}(x)=\arctan x$, strictly increasing with $\Phi_{t_0}'>0$ but not onto $\mathbb{R}$. The hypothesis $m>-\infty$ cannot be dropped either: for $u_0(x)=x^2$, $\Phi_t(x)=x+tx^2$ is not injective for any $t>0$, since $\Phi_t'(x)<0$ for $x<-1/(2t)$.

3 Breakdown of Classical Solutions

3.1 Blow-up of the derivative and the envelope

By the proof of Theorem 2.1,

$$u_x(x,t)=\frac{u_0'(x_0)}{1+u_0'(x_0)\,t},\qquad x=x_0+u_0(x_0)\,t. \tag{10}$$

The formula holds as long as the classical solution exists (Corollary 2.2, that is, for $t<-1/\inf u_0'$). If $u_0'(x_0)\ge0$ the denominator is at least 1, and u_x remains bounded along that characteristic. If $u_0'(x_0)<0$, the right-hand side, regarded

as a formal expression along the characteristic through x_0, has a vanishing denominator at $t = -1/u_0'(x_0)$; the classical solution itself ceases to exist no later than this time, and strictly earlier unless x_0 minimises u_0' (Proposition 3.1). The characteristics

$$x(x_0, t) = x_0 + u_0(x_0)\, t, \qquad x_0 \in \mathbb{R}, \tag{11}$$

have an envelope where $\partial x/\partial x_0 = 1 + u_0'(x_0)\, t = 0$, that is, where $\Phi_t'(x_0) = 0$; this concerns the family (11), not the solution beyond its first breakdown.

3.2 The breakdown time

Proposition 3.1 (Breakdown time)**.** *Let $u_0 \in C^1(\mathbb{R})$ with $m := \inf_{x_0} u_0'(x_0) \in (-\infty, 0)$, and set*

$$T^* = \frac{-1}{m}. \tag{12}$$

Then a C^1 solution exists on $\mathbb{R} \times [0, T^*)$, with $\sup_x |u_x(\cdot, t)| \to \infty$ as $t \uparrow T^*$, and for every $T > T^*$ no C^1 solution exists on $\mathbb{R} \times [0, T)$. If the infimum is attained at $x_0 = a$, the solution degenerates at time T^* at the points

$$x^* = a + u_0(a)\, T^*, \qquad t^* = T^*. \tag{13}$$

Proof. Since $r \mapsto -1/r$ is strictly increasing on $(-\infty, 0)$ and $u_0'(x_0) \geq m$ everywhere, $-1/u_0'(x_0) \geq T^*$ whenever $u_0'(x_0) < 0$, so $\Phi_t' > 0$ throughout $[0, T^*)$ and Corollary 2.2 gives a classical solution u there. Choosing x_n with $u_0'(x_n) \to m$, (10) gives $u_x \to m/(1 + mt)$ at the point with foot x_n, so $\sup_x |u_x(\cdot, t)| \geq |m|/(1 - |m|t) \to \infty$ as $t \uparrow T^*$.

Now let $T > T^*$ and suppose v is a C^1 solution on $\mathbb{R} \times [0, T)$. Since $m < -1/T$, there is a with $u_0'(a) < -1/T$, that is, $\tau := -1/u_0'(a) < T$. By (6) (the characteristic through $(a', 0)$ has bounded slope, hence exists on all of $[0, T)$), $v(a' + tu_0(a'), t) = u_0(a')$ for every a' and $t < T$. Differentiating in a' at $a' = a$ gives

$$v_x\big(a + tu_0(a), t\big)\, \big(1 + t\, u_0'(a)\big) = u_0'(a),$$

and at $t = \tau$ the left side vanishes while the right side is negative, a contradiction. Finally, if the infimum is attained at a, then along the characteristic from a we have $u_x = m/(1 + mt) \to -\infty$ as $t \uparrow T^*$, at points tending to (x^*, T^*) by (11). □

Remark 3.2. Assume that u_0 is bounded, that the infimum m of u_0' is attained, and that $\{u_0' = m\}$ contains no interval. Then $\Phi_{T^*}' \geq 0$ vanishes only on a set with empty interior, so Φ_{T^*} is strictly increasing; it is onto $\mathbb{R}$ because u_0 is bounded, so $\Phi_{T^*}(x_0) \to \pm\infty$ as $x_0 \to \pm\infty$ (this replaces the argument of Corollary 2.2, which degenerates at $t = T^*$). Thus exactly one characteristic reaches each point at time T^*, and u, bounded by $\sup |u_0|$ by (6), develops at (x^*, T^*) only a gradient catastrophe, $|u_x| \to \infty$. If instead $u_0' \equiv m$ on an interval, then Φ_{T^*} is constant on it and characteristics carrying different values of u_0 meet at one point at time T^*, so one expects a discontinuity to form at T^*; this is not proved here (it follows from the Lax–Oleinik formula, Remark 8.4).

3.3 Example: $u_0 = -\arctan x$

Here $u_0'(x_0) = -1/(1 + x_0^2) \in [-1, 0)$ is minimised at $a = 0$ with $m = -1$, so Proposition 3.1 gives $T^* = 1$ and $x^* = 0$: the singularity first appears at $(x^*, t^*) = (0, 1)$ (Figure 1). Here $\Phi_t'(x_0) = 1 - t/(1 + x_0^2)$ first vanishes at $t = 1$, at $x_0 = 0$. At $t = 1$ the map $\Phi_1(x_0) = x_0 - \arctan x_0$ is strictly increasing but its inverse is not C^1 at 0; for $t > 1$, $\Phi_t' < 0$ on an interval around $x_0 = 0$, so Φ_t is not injective and characteristics cross. Geometrically, the characteristic from x_0 has speed $u_0(x_0) = -\arctan x_0$, so those starting near 0, where u_0 falls fastest, converge on one another first, and the crossing begins at the single point $(0, 1)$ rather than everywhere at once. After that time u would take several values at a point, which is why a notion of solution that allows jumps is needed.

By Remark 8.4 and Theorem 6.7, the entropy solution is given by the Lax–Oleinik formula (35). Here $V(y) = -\big(y \arctan y - \frac{1}{2} \ln(1 + y^2)\big)$ is even, and

$$\partial_y G(x, y, t) = \frac{\Phi_t(y) - x}{t}, \qquad \partial_y^2 G = \frac{\Phi_t'(y)}{t} = \frac{1}{t} - \frac{1}{1 + y^2}, \qquad \Phi_t(y) = y - t \arctan y.$$

$t \leq 1$. Then $\partial_y^2 G > 0$ except possibly at $y = 0$, so $G(x, \cdot, t)$ is strictly convex and its unique minimiser is the unique critical point, the characteristic foot $\Phi_t^{-1}(x)$. Thus the entropy solution is the classical solution (7).

$t > 1$. Now Φ_t is odd, with $\Phi_t' < 0$ exactly on $(-s, s)$, $s = \sqrt{t - 1}$. Fix $x > 0$. Since $G(x, y, t) - G(x, -y, t) = -2xy/t$, no minimiser has $y < 0$, and $y = 0$ is not one, since $\partial_y G(x, 0, t) = -x/t < 0$. On $(0, s]$ we have $\Phi_t \leq 0 < x$, and on (s, ∞) Φ_t increases from a negative value to $+\infty$. Hence $G(x, \cdot, t)$ has exactly one critical point $y_+(x, t) > s$ in $(0, \infty)$,

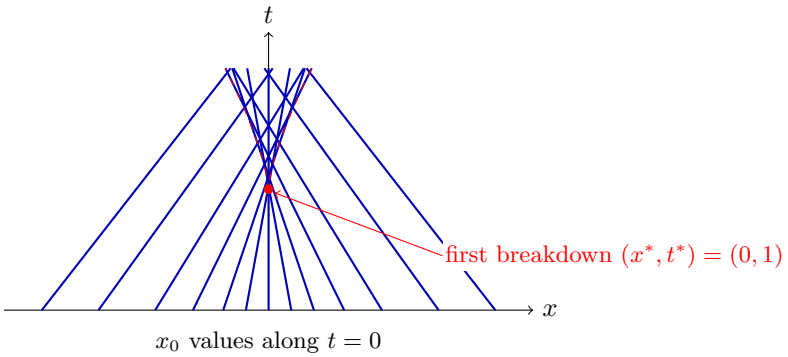


Figure 1: Characteristics $x = x_0 - \arctan(x_0)\, t$ for $u_0 = -\arctan x$, with their envelope (red, dashed), which has a cusp at the first breakdown $(x^*, t^*) = (0, 1)$. Inside the cusp ($t > 1$) three characteristics pass through each point.

with $\Phi_t'(y_+) > 0$; it is the unique minimiser, and $u = (x - y_+)/t = -\arctan y_+$ is smooth in x by the implicit function theorem; by oddness the same holds for $x < 0$. At $x = 0$ the function $G(0, \cdot, t)$ is even with $\partial_y^2 G(0, 0, t) = (1-t)/t < 0$, so 0 is not a minimiser, and the minimisers are $\pm y^*$, where $y^* > 0$ is the unique positive root of $y^* = t \arctan y^*$(the minimum is attained at two symmetric points). Therefore, for $t > 1$, the entropy solution is smooth off $x = 0$ and has exactly one jump, stationary at $x = 0$ with

$$u_\ell = -u_r = \frac{y^*}{t} = \arctan y^* \in (0, \tfrac{\pi}{2}).$$

Moreover $y^* = \sqrt{3(1 - 1/t)}\,(1 + o(1))$ as $t \downarrow 1$ from $\arctan y = y - y^3/3 + O(y^5)$, so $u_\ell \sim \sqrt{3(t-1)}$, and $y^* > s \to \infty$ gives $u_\ell \to \pi/2$ as $t \to \infty$, matching the limits $u_0(\mp\infty) = \pm\pi/2$.

4 Weak Solutions

From here on we assume $u_0 \in L^\infty(\mathbb{R})$, whereas Sections 2–3 allowed unbounded C^1 data; weak solutions are bounded by Definition 4.1, so they do not continue every classical solution considered there.

4.1 The integral formulation

Let $f(u) = u^2/2$. If u is a C^1 solution of $u_t + f(u)_x = 0$ with $u(\cdot, 0) = u_0$ and $\varphi \in C_c^1(\mathbb{R} \times [0, \infty))$, then integration by parts in x and t gives

$$0 = \int_0^\infty \int_{\mathbb{R}} \big(u_t + f(u)_x\big)\varphi \, dx\, dt = -\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\, dx\, dt - \int_{\mathbb{R}} u_0(x)\,\varphi(x, 0)\, dx. \tag{14}$$

Since the right-hand side requires no differentiability of u, it can be used to define solutions of lower regularity.

Definition 4.1 (Weak solution)**.** Let $u_0 \in L^\infty(\mathbb{R})$. A function $u \in L^\infty(\mathbb{R} \times (0, \infty))$ is a *weak solution* of (3) (with data u_0) if

$$\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\, dx\, dt + \int_{\mathbb{R}} u_0(x)\,\varphi(x, 0)\, dx = 0 \tag{15}$$

for every $\varphi \in C_c^1(\mathbb{R} \times [0, \infty))$.

Reversing the integration by parts in (14) and applying the fundamental lemma of the calculus of variations shows that a bounded $u \in C^1(\mathbb{R} \times [0, \infty))$ is a weak solution with data u_0 if and only if it is a classical solution with $u(\cdot, 0) = u_0$.

4.2 Piecewise smooth functions and the Rankine–Hugoniot condition

Definition 4.2 (Piecewise C^1 function)**.** Let $\sigma_1, \dots, \sigma_k : [0, \infty) \to \mathbb{R}$ be C^1 with $\sigma_1(t) < \cdots < \sigma_k(t)$ for $t > 0$ (the curves may start at a common point when $t = 0$). Put $\sigma_0 \equiv -\infty$, $\sigma_{k+1} \equiv +\infty$ and $\Omega_i = \{(x, t) :\ t > 0,\ \sigma_i(t) < x < \sigma_{i+1}(t)\}$ for $i = 0, \dots, k$. A bounded function u is *piecewise C^1 with respect to $\sigma_1, \dots, \sigma_k$* if on each Ω_i it is C^1 and u, u_x, u_t extend continuously to the closure of Ω_i in $\mathbb{R} \times (0, \infty)$, that is, to the points of the boundary curves with $t > 0$. Along $x = \sigma_i(t)$ we write $u_\ell(t)$ and $u_r(t)$ for the limits taken from Ω_{i-1} and Ω_i respectively.

The restriction to finitely many C^1 curves is made so that checking the weak formulation reduces to the PDE on each region together with the Rankine–Hugoniot condition on each curve (Lemma 4.3). This suffices to verify the explicit solutions of Sections 5–7, but it is not a general description of entropy solutions: it excludes, for instance, a gradient catastrophe at an interior time such as $(0,1)$ in Section 3.3, and interacting shocks. General solutions are covered by Definition 4.1 and the existence theory cited in Section 6.

Lemma 4.3 (Boundary identity)**.** *Let u be piecewise C^1 with respect to $\sigma_1,\dots,\sigma_k$ and satisfy $u_t+f(u)_x=0$ classically in each Ω_i. Then for every $\varphi\in C_c^1(\mathbb{R}\times[0,\infty))$ and every $\varepsilon>0$,*

$$\iint_{t>\varepsilon}\big(u\varphi_t+f(u)\varphi_x\big)\,dx\,dt=-\int_{\mathbb{R}}u(x,\varepsilon)\,\varphi(x,\varepsilon)\,dx+\sum_{i=1}^{k}\int_{\varepsilon}^{\infty}\Big[\big(f(u_\ell)-\sigma_i'u_\ell\big)-\big(f(u_r)-\sigma_i'u_r\big)\Big]\varphi(\sigma_i(t),t)\,dt. \quad (16)$$

Proof. On each $\Omega_i^\varepsilon=\Omega_i\cap\{t>\varepsilon\}$ (a Lipschitz domain, bounded after intersecting with $\operatorname{supp}\varphi$) the vector field $\varphi\,(f(u),u)$ (components in the (x,t) order) is C^1 up to the boundary, and its divergence is $u\varphi_t+f(u)\varphi_x$ because $u_t+f(u)_x=0$ there. The divergence theorem gives

$$\iint_{\Omega_i^\varepsilon}(u\varphi_t+f(u)\varphi_x)\,dx\,dt=\oint\varphi\,\big(f(u)\,n_x+u\,n_t\big)\,d\sigma,$$

where $d\sigma\,\mathbf{n}$ equals $(1,-\sigma_{i+1}')\,dt$ on the right edge $x=\sigma_{i+1}(t)$, $(-1,\sigma_i')\,dt$ on the left edge $x=\sigma_i(t)$, and $(0,-1)\,dx$ on the bottom edge $t=\varepsilon$. Summing over i, the bottom edges give $-\int u(x,\varepsilon)\varphi(x,\varepsilon)\,dx$, and the two sides of each curve $x=\sigma_i(t)$ (from Ω_{i-1} on its left, Ω_i on its right) combine into the bracket in (16). □

Theorem 4.4 (Rankine–Hugoniot condition)**.** *Let u be piecewise C^1 with respect to $\sigma_1,\dots,\sigma_k$, and a weak solution of (3) for some data $u_0\in L^\infty$. Then $u_t+f(u)_x=0$ classically in each Ω_i, and along each curve $x=\sigma_i(t)$, $t>0$,*

$$f(u_\ell(t))-f(u_r(t))=\sigma_i'(t)\,\big(u_\ell(t)-u_r(t)\big). \quad (17)$$

In particular, wherever $u_\ell(t)\neq u_r(t)$, and for $f(u)=u^2/2$,

$$\sigma_i'(t)=\frac{f(u_\ell)-f(u_r)}{u_\ell-u_r}=\frac{u_\ell(t)+u_r(t)}{2}. \quad (18)$$

Proof. Test functions φ supported in a single Ω_i give $\iint(u\varphi_t+f(u)\varphi_x)=0$, so $u_t+f(u)_x=0$ in the weak sense there; as u is C^1 on Ω_i, this holds classically, so Lemma 4.3 applies.

Now fix $t_0>0$, an index i, and $\varphi\in C_c^1(\mathbb{R}\times(0,\infty))$ supported in a small ball around $(\sigma_i(t_0),t_0)$ meeting no other curve. Take ε below the support of φ in (16): the left side is 0 by (15) (as $\varphi(\cdot,0)=0$), the bottom term vanishes, and only the curve $x=\sigma_i(t)$ contributes to the sum, so

$$0=\int_0^\infty\Big[\big(f(u_\ell)-\sigma_i'u_\ell\big)-\big(f(u_r)-\sigma_i'u_r\big)\Big]\varphi(\sigma_i(t),t)\,dt.$$

Since $\varphi(\sigma_i(t),t)$ may be any C^1 function of t with small support, the bracket vanishes identically, which is (17). The first form of (18) follows by division, and the second from $f(u_\ell)-f(u_r)=\frac12(u_\ell-u_r)(u_\ell+u_r)$. □

Proposition 4.5 (Piecewise smooth weak solutions)**.** *Let $u_0\in L^\infty(\mathbb{R})$ and let u be piecewise C^1 with respect to $\sigma_1,\dots,\sigma_k$. Suppose*

(i) *$u_t+f(u)_x=0$ classically in each Ω_i;*

(ii) *the jump relation (17) holds along each curve $x=\sigma_i(t)$, $t>0$ (automatic where u is continuous across the curve);*

(iii) *$u(x,t)\to u_0(x)$ as $t\to0^+$ for every $x\notin\{\sigma_1(0),\dots,\sigma_k(0)\}$.*

Then u is a weak solution of (3) with data u_0.

Proof. Fix $\varphi\in C_c^1(\mathbb{R}\times[0,\infty))$ and $\varepsilon>0$. By (i), Lemma 4.3 applies, and by (ii) every term of the sum in (16) vanishes, so

$$\iint_{t>\varepsilon}\big(u\varphi_t+f(u)\varphi_x\big)\,dx\,dt=-\int_{\mathbb{R}}u(x,\varepsilon)\,\varphi(x,\varepsilon)\,dx.$$

Let $\varepsilon\to0^+$. The left side tends to the integral over $t>0$ (bounded integrand, compact support). On the right, $\varphi(\cdot,\varepsilon)\to\varphi(\cdot,0)$ uniformly and, by (iii), $u(\cdot,\varepsilon)\to u_0$ almost everywhere with $|u|\le\|u\|_\infty$, so dominated convergence gives $-\int u_0\varphi(\cdot,0)\,dx$. This is (15); since the argument is run on $t>\varepsilon$, the curves are allowed to meet at $t=0$. □

5 Non-Uniqueness of Weak Solutions

Consider (3) with the Riemann data

$$u_0(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases} \tag{19}$$

Characteristics emanate from $x_0 < 0$ with speed 0 and from $x_0 > 0$ with speed 1, so none covers the wedge $0 < x < t$.

Proposition 5.1 (Non-uniqueness)**.** *The functions*

$$u_A(x,t) = \begin{cases} 0, & x < t/2, \\ 1, & x > t/2, \end{cases} \tag{20}$$

$$u_B(x,t) = \begin{cases} 0, & x \le 0, \\ x/t, & 0 < x < t, \\ 1, & x \ge t, \end{cases} \tag{21}$$

are distinct weak solutions of (3) *with data* (19)*; they differ on the wedge* $0 < x < t$*.*

Proof. Both are covered by Proposition 4.5. For u_A take $k = 1$ and $\sigma_1(t) = t/2$: u_A is constant on each side and the jump moves at the Rankine–Hugoniot speed $(0+1)/2 = \frac{1}{2}$ of (18). For u_B take $k = 2$, $\sigma_1 \equiv 0$, $\sigma_2(t) = t$: u_B is constant on the outer regions, solves the PDE on $0 < x < t$ since $(u_B)_t + u_B(u_B)_x = -x/t^2 + (x/t)(1/t) = 0$, and is continuous across both curves, so (17) holds trivially. In both cases $u(\cdot,t) \to u_0$ at every $x \neq 0$. □

Remark 5.2 (An infinite family). More generally, for $\alpha \in \mathbb{R}$ the function equal to 0 for $x < \frac{\alpha}{2}t$, to α for $\frac{\alpha}{2}t < x < \frac{1+\alpha}{2}t$ and to 1 for $x > \frac{1+\alpha}{2}t$ is a weak solution with data (19) by Proposition 4.5 (its two jumps move at their Rankine–Hugoniot speeds $\alpha/2 < (1+\alpha)/2$); we denote it $u^{(\alpha)}$, and $u^{(0)} = u^{(1)} = u_A$.

6 The Entropy Condition

6.1 The Lax shock condition

Definition 6.1 (Lax shock condition)**.** Let u be a weak solution of a scalar conservation law $u_t + f(u)_x = 0$ with $f \in C^2$, $f'' > 0$, and suppose u has a jump discontinuity along a C^1 curve $x = \sigma(t)$ with left and right limits $u_\ell(t)$, $u_r(t)$. The jump satisfies the *Lax shock condition* if

$$f'(u_\ell(t)) > \sigma'(t) > f'(u_r(t)) \tag{22}$$

for all t.

For strictly convex f the Rankine–Hugoniot speed $(f(u_\ell) - f(u_r))/(u_\ell - u_r)$ lies strictly between $f'(u_\ell)$ and $f'(u_r)$, in the order of u_ℓ, u_r (mean value theorem), so (22) holds if and only if $u_\ell > u_r$. For $f(u) = u^2/2$ and $\sigma' = (u_\ell + u_r)/2$, (22) reduces to

$$u_\ell > \frac{u_\ell + u_r}{2} > u_r \quad \Longleftrightarrow \quad u_\ell > u_r. \tag{23}$$

Thus characteristics on both sides run into the shock. The Lax condition is local: it tests one jump at a time and says nothing about the smooth parts of the solution, whereas the Oleinik condition of the next subsection is global in x and does (Remark 6.5). It also depends on convexity: for non-convex f the Lax inequalities no longer characterise admissible shocks, and one passes to Oleinik's condition E or to Kružkov's entropy inequalities [8, 10], which we do not pursue. Since the Lax condition constrains jumps only, its derivation from viscosity, giving $u_\ell > u_r$, is deferred to Proposition 8.6.

6.2 Entropy solutions: the Oleinik condition

Definition 6.2 (Entropy solution)**.** A weak solution $u \in L^\infty$ of (3) is an *entropy solution* if

$$u(x+z,t) - u(x,t) \;\le\; \frac{z}{t} \qquad \text{for every } z > 0 \text{ and a.e. } (x,t) \in \mathbb{R} \times (0,\infty). \tag{24}$$

The one-sided Lipschitz bound (24) is due to Oleinik [13]. It prevents $u(\cdot,t)$ from increasing at a rate exceeding $1/t$, which is what excludes upward jumps (Lemma 6.4).

Remark 6.3 (Entropy inequalities). The name refers to the following mechanism. Let η be convex with entropy flux q, $q' = \eta' f'$. A smooth solution of (1) satisfies

$$\eta(u)_t + q(u)_x = \nu\,\eta(u)_{xx} - \nu\,\eta''(u)\,u_x^2 \;\le\; \nu\,\eta(u)_{xx},$$

so that, as $\nu \to 0$, the limit satisfies $\eta(u)_t + q(u)_x \le 0$ in the sense of distributions: entropy is dissipated across shocks, never produced. For Burgers' equation, requiring this for all convex η is equivalent to (24) [8, 10]; we use (24) because it is the form needed for uniqueness, and do not prove the equivalence.

Lemma 6.4 (Oleinik implies Lax)**.** *Let u be a piecewise C^1 entropy solution of Burgers' equation. Then every jump of u satisfies the Lax condition* (22)*, equivalently $u_\ell > u_r$.*

Proof. Fix a curve $x = \sigma(t)$ and $t_0 > 0$ with $u_\ell(t_0) \neq u_r(t_0)$. For $z > 0$ small enough that $[\sigma(t_0) - z, \sigma(t_0) + z]$ meets no other curve, there is a neighbourhood N of $(\sigma(t_0), t_0)$ in which every pair $(x,t), (x+z,t)$ with $x < \sigma(t) < x + z$ lies on opposite sides of the curve $x = \sigma(t)$ only, where u is C^1. Both sides of (24) are continuous on the open set $\{(x,t) \in N : x < \sigma(t) < x + z\}$, so the a.e. inequality holds at every point of it, in particular at $(\sigma(t_0) - z/2, t_0)$. Letting $z \to 0$, the left side tends to $u_r(t_0) - u_\ell(t_0)$ and the right side to 0. Hence $u_r \le u_\ell$ at t_0, and since the limits differ, $u_\ell > u_r$. □

Remark 6.5. For a piecewise C^1 solution with locally finitely many jumps, (24) holds if and only if each jump satisfies $u_\ell > u_r$ and $u_x \le 1/t$ on the smooth pieces. The Lax condition is thus a local admissibility condition, pointwise on each jump, and is the local form of (24) at jumps for piecewise C^1 solutions. The Oleinik bound is global in x: it also constrains the smooth parts ($u_x \le 1/t$), and it is meaningful for general L^∞ weak solutions, for which no jump structure is available. It is the bound used in the uniqueness proof below. Lemma 6.4 is used only to test the explicit piecewise C^1 solutions; the uniqueness theorem (Theorem 6.7) assumes (24) directly and does not pass through the Lax condition, so there is no circularity.

Proposition 6.6. *For the Riemann data* (19) *($u_\ell = 0 < u_r = 1$), the rarefaction solution u_B of* (21) *is an entropy solution, while the shock solution u_A of* (20) *and every $u^{(\alpha)}$ of Remark 5.2 are not.*

Proof. u_B is entropy. Write $u_B(x,t) = \min\{\max\{x/t, 0\}, 1\}$, the composition of $x \mapsto x/t$ with the 1-Lipschitz function $y \mapsto \min\{\max\{y, 0\}, 1\}$. Hence $u_B(\cdot, t)$ is $(1/t)$-Lipschitz and $u_B(x+z,t) - u_B(x,t) \le z/t$, which is (24).

u_A and $u^{(\alpha)}$ are not. Each contains an upward jump ($0 \to 1$ for u_A; $0 \to \alpha$ if $\alpha > 0$ and $\alpha \to 1$ if $\alpha \le 0$ for $u^{(\alpha)}$), which Lemma 6.4 excludes. (Directly, for u_A and $z < t$ the increment equals $1 > z/t$ for every $x \in (t/2 - z, t/2)$.) □

6.3 Uniqueness

Theorem 6.7 (Oleinik; [5, §3.4.3])**.** *For $u_0 \in L^\infty(\mathbb{R})$ and $f(u) = u^2/2$, the initial value problem $u_t + f(u)_x = 0$, $u(\cdot, 0) = u_0$, has at most one entropy solution in the sense of Definition 6.2, up to a set of measure zero. An entropy solution exists: the Lax–Oleinik solution [5, §3.4.2] is a weak solution satisfying* (24) *(Remark 8.4).*

(Evans proves uniqueness under the weaker bound $u(x+z,t) - u(x,t) \le C(1 + 1/t)\,z$, valid for general uniformly convex fluxes, which every solution of (24) satisfies with $C = 1$.) The proof in [5, §3.4.3] proceeds by duality. The difference $w = u_1 - u_2$ of two entropy solutions with the same data satisfies $\iint w\,(\varphi_t + b\,\varphi_x)\,dx\,dt = 0$ for all test functions φ, where $b = \frac{1}{2}(u_1 + u_2)$, so $w = 0$ follows if the adjoint transport problem $v_t + b\,v_x = \psi$ can be solved backward in time for arbitrary ψ. This is where the entropy condition enters: the one-sided bound gives $b_x \le C/t$, which keeps the dual solutions of bounded variation for $t > 0$, whereas without such a bound the dual problem is ill-posed, as it must be in view of Section 5. The estimates are given in the reference. Heuristically, the bound forbids the mechanism behind u_A: an upward jump emits characteristics from a single point, so that the solution there is a free choice, whereas under (24) characteristics can only run into discontinuities, never out of them, and every value is determined by the data. Kružkov's formulation via entropy inequalities [8] instead yields an L^1-contraction and covers non-convex fluxes.

7 The Riemann Problem

The entropy solution is now determined for arbitrary constant states $u_\ell, u_r \in \mathbb{R}$, that is, for the data

$$u_0(x) = \begin{cases} u_\ell, & x < 0, \\ u_r, & x > 0. \end{cases} \tag{25}$$

7.1 The rarefaction fan

Seek self-similar solutions $u(x,t) = g(\xi)$, $\xi = x/t$. Then

$$u_t + u\,u_x = -\frac{\xi}{t}g'(\xi) + \frac{g(\xi)}{t}g'(\xi) = \frac{g'(\xi)}{t}\big(g(\xi) - \xi\big) = 0, \tag{26}$$

so on each interval of ξ either $g'(\xi) = 0$ (a constant state) or $g(\xi) = \xi$ (the fan $u = x/t$), which, being increasing in x, can only connect states with $u_\ell < u_r$.

7.2 The complete solution

Theorem 7.1 (Entropy solution of the Riemann problem)**.** *The unique entropy solution of* (3) *with data* (25) *is:*

(a) *If $u_\ell > u_r$ (shock case),*

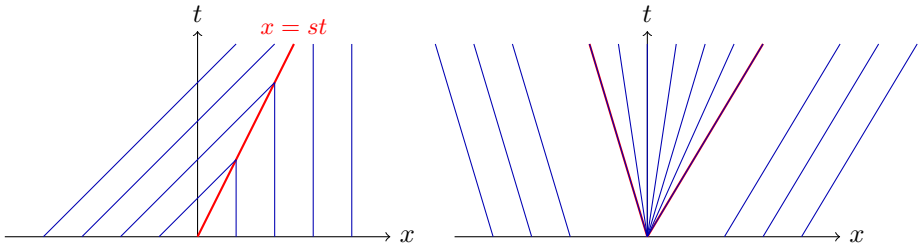
$$u(x,t) = \begin{cases} u_\ell, & x < st, \\ u_r, & x > st, \end{cases} \qquad s = \frac{u_\ell + u_r}{2}. \tag{27}$$

(b) *If $u_\ell \le u_r$ (rarefaction case),*

$$u(x,t) = \begin{cases} u_\ell, & x \le u_\ell t, \\ x/t, & u_\ell t < x < u_r t, \\ u_r, & x \ge u_r t. \end{cases} \tag{28}$$

Proof. (a) As in the proof of Proposition 5.1, with $\sigma_1(t) = st$ and $s = (u_\ell + u_r)/2$ the Rankine–Hugoniot speed, u is a weak solution. It is also an entropy solution, since the only possible increments of $u(\cdot,t)$ are 0 and $u_r - u_\ell < 0$, whence $u(x+z,t) - u(x,t) \le 0 \le z/t$; by Theorem 6.7 it is the only one. Its characteristics are shown in Figure 2(a).

(b) The case $u_\ell = u_r$ being trivial, let $u_\ell < u_r$. As in Proposition 5.1, with $\sigma_1(t) = u_\ell t$ and $\sigma_2(t) = u_r t$, u is a weak solution: it solves the PDE on the fan by Section 7.1 and is continuous across both curves. Moreover $u = \min\{\max\{x/t, u_\ell\}, u_r\}$ is $(1/t)$-Lipschitz in x, so (24) holds, and uniqueness follows from Theorem 6.7 (see Figure 2(b)). □



(a) shock: $u_\ell > u_r$ (b) rarefaction: $u_\ell < u_r$

Figure 2: Characteristics for the two cases of Theorem 7.1. (a) $u_\ell = 1$, $u_r = 0$, $s = 1/2$: characteristics on both sides run into the shock $x = st$. (b) Characteristics fan out from the origin.

8 Extensions

Section 5 showed that conservation alone leaves infinitely many weak solutions, and the Oleinik condition restores uniqueness, but it does not explain *why* nature, or a physical model with a small viscosity, picks that solution. Can the selected solution be derived from the underlying viscous equation? This section tests the entropy condition against the two standard selection mechanisms: vanishing viscosity (Subsections 8.1–8.3) and monotone numerical approximation (Subsection 8.4).

8.1 The Cole–Hopf transformation

Theorem 8.1 (Cole–Hopf transformation [3, 7])**.** *Let $u \in C^{2,1}(\mathbb{R}\times(0,T))$ solve $u_t + u\,u_x = \nu\,u_{xx}$, with u and u_x continuous on $\mathbb{R}\times[0,T)$ (so that c below is defined), and set*

$$\hat\psi(x,t) = -\frac{1}{2\nu}\int_0^x u(y,t)\,dy, \qquad c(t) = \int_0^t \Big(\frac{u(0,\tau)^2}{4\nu} - \frac{u_x(0,\tau)}{2}\Big)d\tau, \qquad \theta(x,t) = \exp\big(\hat\psi(x,t) + c(t)\big) > 0. \tag{29}$$

Then θ solves the heat equation $\theta_t = \nu\,\theta_{xx}$, and $u = -2\nu\,\theta_x/\theta$.

Proof. Since c depends on t only, $\theta_x/\theta = \hat\psi_x = -u/(2\nu)$, which gives $u = -2\nu\theta_x/\theta$. For the heat equation, let $w = \hat\psi_t - \nu\hat\psi_{xx} - \nu\hat\psi_x^2$. Using $\hat\psi_x = -u/(2\nu)$, $\hat\psi_{xx} = -u_x/(2\nu)$, $\hat\psi_{xt} = -u_t/(2\nu)$ and $\hat\psi_{xxx} = -u_{xx}/(2\nu)$,

$$w_x = \hat\psi_{xt} - \nu\hat\psi_{xxx} - 2\nu\hat\psi_x\hat\psi_{xx} = -\frac{1}{2\nu}\big(u_t + u\,u_x - \nu\,u_{xx}\big) = 0. \tag{30}$$

Hence w depends on t only and may be evaluated at $x = 0$, where $\hat\psi(0,t) = 0$ gives $\hat\psi_t(0,t) = 0$:

$$w(t) = w(0,t) = \frac{u_x(0,t)}{2} - \frac{u(0,t)^2}{4\nu} =: C(t). \tag{31}$$

With $\psi = \hat\psi + c$ we have $c' = -C$, so $\psi_t - \nu\psi_{xx} - \nu\psi_x^2 = C + c' = 0$. Since $\theta = e^{\psi}$ satisfies $\theta_t = \theta\psi_t$ and $\theta_{xx} = \theta(\psi_{xx} + \psi_x^2)$, this is $\theta_t = \nu\theta_{xx}$. (The factor $e^{c(t)}$ depends on t only, so it affects neither $u = -2\nu\theta_x/\theta$ nor the initial data, as $c(0) = 0$; it cannot be omitted, since $e^{\hat\psi}$ satisfies $\partial_t\theta = \nu\theta_{xx} + C(t)\theta$.) □

8.2 Exact solution via the heat kernel

By Theorem 8.1, a classical solution of (1) that is bounded on $\mathbb{R}\times[0,T']$ for each $T' < T$ (this is what makes θ obey the growth bound below) corresponds to a solution of the heat equation with initial data

$$\theta_0(x) = \exp\left(-\frac{1}{2\nu}\int_0^x u_0(y)\,dy\right) \le e^{\|u_0\|_\infty |x|/(2\nu)}. \tag{32}$$

The Cauchy problem for the heat equation has a unique solution in the class of at most $e^{a|x|^2}$ growth [5, §2.3.3], which contains this data; it is given by

$$\theta(x,t) = \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi\nu t}}\,\exp\left(-\frac{(x-y)^2}{4\nu t}\right)\theta_0(y)\,dy. \tag{33}$$

Substituting (32)–(33) into $u = -2\nu\,\theta_x/\theta$ and differentiating under the integral gives the *Cole–Hopf formula*

$$u(x,t) = \frac{\displaystyle\int_{\mathbb{R}} \frac{x-y}{t}\,\exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}{\displaystyle\int_{\mathbb{R}} \exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}, \qquad G(x,y,t) = \int_0^y u_0(r)\,dr + \frac{(x-y)^2}{2t}. \tag{34}$$

Corollary 8.2 (Global smoothness; Hopf [7])**.** *For every bounded measurable u_0 and every $\nu > 0$, formula* (34) *defines a C^∞ solution $u^{(\nu)}$ of* (1) *on $\mathbb{R}\times(0,\infty)$. Moreover $\|u^{(\nu)}\|_\infty \le \|u_0\|_\infty$, $u^{(\nu)}(\cdot,t) \to u_0$ in L^1_{loc} and at every continuity point of u_0 as $t \to 0^+$, and $u^{(\nu)}$ is the unique bounded solution with these properties.*

For $t > 0$, θ in (33) is smooth and positive, and for any such solution θ of the heat equation $u = -2\nu\theta_x/\theta$ solves (1): with $\psi = \ln\theta$, $\psi_t - \nu\psi_{xx} - \nu\psi_x^2 = 0$, and differentiating in x gives $(-u/2\nu)_t - \nu(-u/2\nu)_{xx} - \nu\big((u/2\nu)^2\big)_x = 0$, i.e. $u_t + u\,u_x = \nu u_{xx}$. This proves the first assertion for merely measurable u_0 (Theorem 8.1 is not needed in this direction), while the bound, the sense in which the data are attained, and uniqueness are taken from Hopf [7].

8.3 The vanishing viscosity limit

Theorem 8.3 (Vanishing viscosity limit [7])**.** *Let $u_0 \in L^\infty(\mathbb{R})$, let $u^{(\nu)}$ be the solution of* (1) *with data u_0 given by* (34)*, and let u be the entropy solution of* (2) *with the same data (Definition 6.2; for Riemann data, Theorem 7.1). Then $u^{(\nu)} \to u$ pointwise almost everywhere and in L^1_{loc} as $\nu \to 0^+$.*

Hopf's proof [7] applies Laplace's method to (34) and shows that $u^{(\nu)}$ converges to the Lax–Oleinik solution (35). Because his work predates Oleinik's condition, the identification of this limit with u combines the theorem with Remark 8.4 below, which shows that the Lax–Oleinik solution satisfies (24), and with Theorem 6.7. In short: the viscous solutions converge to the Lax–Oleinik solution (Hopf); that solution is a weak solution obeying (24) (Remark 8.4); and weak solutions obeying (24) are unique, so the limit is u.

Remark 8.4 (The Lax–Oleinik formula). As $\nu \to 0$ the weight $\exp(-G/2\nu)$ in (34) concentrates at the minimisers of $y \mapsto G(x,y,t)$. Where the minimiser $y^*(x,t)$ is unique, the limit is

$$u(x,t) = \frac{x - y^*(x,t)}{t}, \qquad y^*(x,t) = \arg\min_y \Big\{ V(y) + \frac{(x-y)^2}{2t} \Big\}, \quad V(y) = \int_0^y u_0, \tag{35}$$

the *Lax–Oleinik formula* [9, 13], [5, §3.4.2]; a jump occurs where two minimisers tie. Since $x \mapsto y^*(x,t)$ is nondecreasing [5, §3.4.2], for $z > 0$

$$u(x+z,t) - u(x,t) = \frac{z - \big(y^*(x+z,t) - y^*(x,t)\big)}{t} \;\le\; \frac{z}{t},$$

which is (24). As (35) also defines a weak solution [5, §3.4.2], the Lax–Oleinik solution is an entropy solution in the sense of Definition 6.2. The same bound shows $u(x+,t) - u(x-,t) \le 0$ wherever one-sided limits exist: every jump is downward. The minimiser is unique for almost every (x,t) [5, §3.4.2], so (35) defines u a.e.

Remark 8.5 (Riemann data and the Lax–Oleinik formula). For the data (25), $V(y) = u_\ell y$ for $y < 0$ and $V(y) = u_r y$ for $y > 0$, so $G(y) = V(y) + (x-y)^2/2t$ has a kink at $y = 0$ and critical points $y_\ell = x - u_\ell t$ (admissible iff $x \le u_\ell t$) and $y_r = x - u_r t$ (iff $x \ge u_r t$). If $u_\ell < u_r$ and $u_\ell t < x < u_r t$, only the kink is a minimiser, giving the fan $u = x/t$. If $u_\ell > u_r$ and $u_r t < x < u_\ell t$, both y_ℓ, y_r are admissible, the kink is a local maximum, and $G(y_\ell) - G(y_r) = (u_\ell - u_r)\big(x - \frac{u_\ell + u_r}{2} t\big)$, so the minimum switches at $x = \frac{u_\ell + u_r}{2} t$: the Rankine–Hugoniot speed is the position at which two minimisers tie.

8.3.1 Viscous shock profiles

Proposition 8.6 (Viscous shock profiles)**.** *Let $u_\ell \ne u_r$ and $\nu > 0$. Equation (1) has a travelling-wave solution $u(x,t) = \phi(x - st)$, with $\phi \in C^2$ bounded, $\phi(\eta) \to u_\ell$ as $\eta \to -\infty$, $\phi(\eta) \to u_r$ as $\eta \to +\infty$ and $\phi' \to 0$ as $\eta \to \pm\infty$, if and only if $u_\ell > u_r$. In that case $s = (u_\ell + u_r)/2$ and, with $\Delta = u_\ell - u_r$, ϕ is unique up to translation $\eta \mapsto \eta + \eta_0$; the translate with $\phi(0) = s$ is*

$$\phi(\eta) = s - \frac{\Delta}{2} \tanh\Big(\frac{\Delta\,\eta}{4\nu}\Big). \tag{36}$$

Moreover $\|\phi - (s - \frac{\Delta}{2}\operatorname{sgn}\eta)\|_{L^1(\mathbb{R})} = 4\ln 2\,\nu$, independent of s and Δ.

Proof. Substituting $u = \phi(x - st)$ gives $-s\phi' + \phi\phi' = \nu\phi''$, and integrating once from $-\infty$ (where $\phi' \to 0$),

$$\nu\phi' = \tfrac{1}{2}\phi^2 - s\phi - K, \qquad K = \tfrac{1}{2}u_\ell^2 - s u_\ell.$$

Letting $\eta \to +\infty$ shows that the right-hand side must also vanish at u_r, so u_ℓ, u_r are the two roots of the quadratic: $s = (u_\ell + u_r)/2$ (the Rankine–Hugoniot speed) and

$$\nu\phi' = \tfrac{1}{2}(\phi - u_\ell)(\phi - u_r). \tag{37}$$

By uniqueness for this autonomous ODE, a nonconstant solution cannot cross either equilibrium u_ℓ, u_r, so a connecting ϕ lies strictly between them. On $(\min(u_\ell,u_r), \max(u_\ell,u_r))$ the right side of (37) is negative, so ϕ is decreasing. If $u_\ell < u_r$ it would have to increase from u_ℓ to u_r, a contradiction; if $u_\ell > u_r$, (37) is solved by (36); any other solution is a translate, by uniqueness for the ODE. For the L^1 distance, by oddness and $\int_0^\infty (1 - \tanh\zeta)\,d\zeta = \ln 2$,

$$\int_{\mathbb{R}} \tfrac{\Delta}{2} \Big| \tanh\tfrac{\Delta\eta}{4\nu} - \operatorname{sgn}\eta \Big|\, d\eta = \Delta \int_0^\infty \Big(1 - \tanh\tfrac{\Delta\eta}{4\nu}\Big) d\eta = \Delta \cdot \frac{4\nu}{\Delta} \ln 2 = 4\ln 2\,\nu.$$

□

Proposition 8.6 determines which isolated jumps admit a viscous connecting profile, namely exactly those with $u_\ell > u_r$, of thickness $O(\nu/\Delta)$; this motivates the Lax condition (23) but does not by itself select a solution of the initial-value problem. That selection is the role of Theorem 8.3, which together with Proposition 6.6 excludes u_A of Section 5. As a check, for the shock data $(1, 0)$ the closed-form Cole–Hopf solution (`cole_hopf_riemann.py` [12]) lies at $L^1(-2,2)$ distance $2.773\,\nu$ from the entropy solution at $t = 1$ for $\nu \le 10^{-3}$, matching the layer value $4\ln 2\,\nu$ above.

8.4 Godunov's method

8.4.1 The scheme

Let $\mathbb{R}$ be divided into cells $[x_{j-1/2}, x_{j+1/2}]$ of width Δx, with $t^n = n\Delta t$ and cell averages $\bar u_j^n = \Delta x^{-1} \int_{x_{j-1/2}}^{x_{j+1/2}} u(x,t^n)\,dx$ approximated by U_j^n. Integrating $u_t + f(u)_x = 0$ over a cell and a time step gives

$$\bar u_j^{n+1} = \bar u_j^n - \frac{\Delta t}{\Delta x}\Big(F_{j+1/2}^n - F_{j-1/2}^n\Big), \qquad F_{j+1/2}^n = \frac{1}{\Delta t}\int_{t^n}^{t^{n+1}} f\big(u(x_{j+1/2},t)\big)\,dt, \tag{38}$$

for the exact cell averages $\bar{u}_j^n$. The scheme replaces $F_{j+1/2}^n$ by a numerical flux $\widehat{f}_{j+1/2}$ depending on U_j^n, U_{j+1}^n, giving $U_j^{n+1} = U_j^n - \frac{\Delta t}{\Delta x}(\widehat{f}_{j+1/2} - \widehat{f}_{j-1/2})$. Godunov's method [6] takes $\widehat{f}_{j+1/2}$ to be the flux at $x/t = 0$ of the exact Riemann solution with data (U_j^n, U_{j+1}^n) (Theorem 7.1):

$$\widehat{f}_{j+1/2} = f\big(\omega(0;\, U_j^n, U_{j+1}^n)\big), \tag{39}$$

where $\omega(\xi; u_\ell, u_r)$ is the entropy solution with data (u_ℓ, u_r) at $\xi = x/t$. By (27)–(28),

$$\omega(0; u_\ell, u_r) = \begin{cases} u_\ell, & u_\ell > u_r,\ \frac{u_\ell + u_r}{2} > 0, \\ u_r, & u_\ell > u_r,\ \frac{u_\ell + u_r}{2} < 0, \\ u_\ell, & u_\ell \le u_r,\ u_\ell \ge 0, \\ u_r, & u_\ell \le u_r,\ u_r \le 0, \\ 0, & u_\ell < 0 < u_r, \end{cases} \tag{40}$$

the last case corresponding to $\xi = 0$ lying inside the fan. If $u_\ell = -u_r$ the value $\omega(0)$ is ambiguous, but $f(u_\ell) = f(u_r)$, so (39) is unaffected.

8.4.2 Stability and convergence

Monotonicity is the property that makes the scheme entropy-respecting: a scheme is monotone if raising the data U_j^n never lowers U_i^{n+1}, and monotone conservative schemes cannot create new extrema or oscillations, so they behave like a viscous regularisation, whereas consistency and conservation alone do not (Section 8.4.5). The scheme is monotone provided the CFL condition

$$\Delta t \le \frac{\Delta x}{\max_j |U_j^n|} \tag{41}$$

holds, which prevents waves from neighbouring interfaces from reaching $x_{j+1/2}$ within one step [10, Ch. 4].

Theorem 8.7 ([4, Theorem 1 and §1])**.** *Let $u_0 \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$ with $a \le u_0 \le b$ a.e., and let $\lambda = \Delta t/\Delta x$ be fixed with $\lambda \max_{a\le u\le b} |f'(u)| \le 1$. Let u^Δ be the piecewise constant function equal to U_j^n on $[x_{j-1/2}, x_{j+1/2}) \times [t^n, t^{n+1})$, where U_j^0 is the cell average of u_0 and the U_j^n are the Godunov approximations. Then for every $T > 0$*

$$\lim_{\Delta t \to 0} \sup_{0 \le t \le T} \int_{\mathbb{R}} \big|u^\Delta(x,t) - u(x,t)\big|\, dx = 0,$$

where u is the Kružkov entropy solution [8] with data u_0.

The theorem holds for any consistent conservation-form scheme that is monotone on $[a, b]$ with Lipschitz numerical flux, and Godunov's scheme is such a scheme for $f'' > 0$ ([4, §1]); it gives no rate. For Burgers' equation the Kružkov solution is the Oleinik solution of Theorem 6.7, for the following reason. By Theorem 8.3 the Oleinik solution u is the L^1_{loc} limit of viscous solutions $u^{(\nu)}$. These satisfy $|u^{(\nu)} - k|_t + (\operatorname{sgn}(u^{(\nu)} - k)(f(u^{(\nu)}) - f(k)))_x \le \nu |u^{(\nu)} - k|_{xx}$ for every $k \in \mathbb{R}$ (Remark 6.3 with $\eta(u) = |u - k|$, by smooth approximation), and letting $\nu \to 0$ gives the same inequality with $\nu = 0$, which is Kružkov's definition of an entropy solution. Hence u is a Kružkov solution, and by Kružkov's uniqueness theorem [8] it is the only one.

8.4.3 Numerical results

Since the Riemann data used below are not in $L^1(\mathbb{R})$, we replace them by 0 outside $[-5, 5]$; as $|u| \le 1$ and the numerical domain of dependence over $t \le 1$ has half-width $t/\lambda < 2.23$, this changes neither the exact nor the computed solution on $[-2, 2] \times [0, 1]$. Since $\max_j |U_j^n| = 1$ in the experiments, $\Delta t = 0.45\, \Delta x / \max_j |U_j^n|$ gives the constant $\lambda = 0.45$ of the theorem (the final step, shortened to end at $t = 1$, has smaller λ). The scheme was implemented in `godunov.py` [12] on $x \in [-2, 2]$ with N cells, cell-averaged initial data and outflow boundary conditions. The rarefaction data $(0, 1)$ and shock data $(1, 0)$ were run to $t = 1$, with error $\sum_j \int_{I_j} |U_j - u_{\mathrm{exact}}(x, 1)|\, dx$ against the solution of Theorem 7.1 (not cell-averaged, so a cell containing the shock is scored against the discontinuity itself). The code checks that the case-based flux (39) agrees with the closed form on random state pairs, that mass is conserved up to boundary flux, and that enlarging the domain to $[-4, 4]$ changes the values on $[-2, 2]$ by at most 3×10^{-14}. Figure 3 shows $N = 400$: the computed shock lies at $x = st = 0.5$ and the fan has slope $1/t$, with no oscillation.

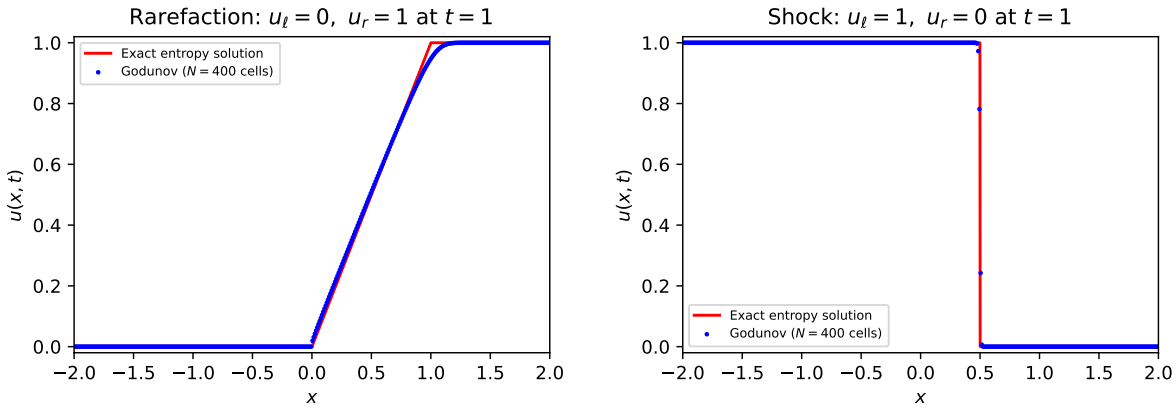


Figure 3: Godunov scheme (dots, $N = 400$) against the exact entropy solution (solid) at $t = 1$. Left: rarefaction data $(0, 1)$. Right: shock data $(1, 0)$. Code: [12].

N	Δx	L^1 (rarefaction)	order	L^1 (shock)	order
200	2.0×10^{-2}	3.05×10^{-2}	—	9.91×10^{-3}	—
400	1.0×10^{-2}	1.82×10^{-2}	0.74	4.98×10^{-3}	0.99
800	5.0×10^{-3}	1.07×10^{-2}	0.77	2.50×10^{-3}	0.99
1600	2.5×10^{-3}	6.14×10^{-3}	0.80	1.24×10^{-3}	1.01
3200	1.25×10^{-3}	3.48×10^{-3}	0.82	6.23×10^{-4}	0.99
6400	6.25×10^{-4}	1.95×10^{-3}	0.84	3.13×10^{-4}	0.99

Table 1: L^1 errors at $t = 1$ and observed orders $\log_2(e_{N/2}/e_N)$. All N are multiples of 8, so the shock lies on a cell interface. Code: [12].

Shock. Here $E/\Delta x$ is the shock error measured in cell widths, so a value c means the computed profile differs from the sharp shock by an amount equal to c cells' worth of mass. The shock $x = st = 0.5$ lies on a cell interface iff $8 \mid N$, which matters because otherwise the cell containing it holds a mixed initial average and the initial error depends on where the discontinuity falls inside that cell; for such N the ratio $E/\Delta x$ lies between 0.495 and 0.501 and the observed order is ≈ 1 (Table 1). On misaligned grids ($N = 97, \dots, 103$) it ranges from $0.55\,\Delta x$ to $0.66\,\Delta x$, so orders computed between misaligned and aligned resolutions are unreliable (from $N = 100$ to 200 one would obtain 1.40); this is why the table uses $N = 200 \cdot 2^k$.

Rarefaction. The observed orders $0.74, 0.77, 0.80, 0.82, 0.84$ are still increasing, so these experiments do not establish a first-order rate. In fact $E/\Delta x$ grows by 0.30 to 0.34 per halving of Δx, roughly linearly in $\ln(1/\Delta x)$, which is consistent with, but does not prove, an error of the form $\Delta x\,(c + c' \ln(1/\Delta x))$, that is, first order up to a logarithm. The error is concentrated near the fan but not at its corners: the fan interior $0.2 < x < 0.8$ carries about a third of it, so the rate has not stabilised.

8.4.4 Numerical diffusion as vanishing viscosity

For the shock data the scheme is monotone, so by the discrete maximum principle all states remain in $[0, 1]$. All characteristic speeds are then nonnegative, so the Godunov flux is the upwind flux $f(U_j^n)$, and the scheme reads $U_j^{n+1} = U_j^n - \lambda(f(U_j^n) - f(U_{j-1}^n))$, $\lambda = \Delta t/\Delta x$. The mechanism is that upwinding averages information from one side, and the leading error of such an average is a diffusion term. Taylor expansion gives the modified equation

$$u_t + u\,u_x = \frac{\Delta x}{2}\big(u(1 - \lambda u)\,u_x\big)_x + O(\Delta x^2), \tag{42}$$

a Burgers-type equation in conservative form, with state-dependent viscosity $\nu(u) = \frac{\Delta x}{2} u(1 - \lambda u)$, nonnegative exactly under (41). In the scaled variable $\eta = (x - t/2)/\Delta x$, its travelling shock $1 \to 0$ of speed $\frac{1}{2}$ satisfies $\frac{1}{2}u(1 - \lambda u)u_\eta = \frac{1}{2}u(u - 1)$, that is,

$$\frac{du}{d\eta} = \frac{u - 1}{1 - \lambda u}, \qquad \eta(u) = \eta_0 + (1 - \lambda)\ln(1 - u) - \lambda u. \tag{43}$$

The profile decays exponentially to 1 as $\eta \to -\infty$ and reaches 0 at a finite η_0, which conservation of mass fixes as $\eta_0 = 1 - \lambda/2$. The L^1 distance from the sharp shock is then

$$\frac{\|u - u_{\text{exact}}\|_{L^1}}{\Delta x} = 2\int_0^{u_0} \frac{u(1-\lambda u)}{1-u}\,du, \qquad (1-\lambda)\ln(1-u_0) - \lambda u_0 = -\Big(1 - \frac{\lambda}{2}\Big). \tag{44}$$

Table 2 compares this with the measured shock error on aligned grids, where λ equals the CFL number since $\max|U| = 1$ (`numerical_viscosity.py` [12]).

CFL $= \lambda$	measured ($N = 1600$)	predicted	(pred.−meas.)/meas.
0.10	0.626	0.686	+9.4%
0.45	0.496	0.511	+3.1%
0.80	0.352	0.342	−3.0%
0.95	0.300	0.272	−9.3%

Table 2: Shock L^1 error divided by Δx: measured (grids with $8 \mid N$) against the modified-equation prediction (code: [12]).

The prediction is within 10% throughout (the CFL 0.95 value is not yet asymptotic: 0.334 at $N = 400$). Equating the error with $4\ln 2\,\nu_{\text{eff}}$ (Proposition 8.6, $s = \frac{1}{2}$, $\Delta = 1$) defines an effective viscosity: the measured $0.50\,\Delta x$ gives $\nu_{\text{eff}} \approx 0.179\,\Delta x$ and the modified-equation profile gives $0.184\,\Delta x$, so the effective viscosity vanishes with the mesh, the numerical analogue of the mechanism behind Theorem 8.3. That theorem concerns constant physical viscosity and does not itself give convergence of the scheme, whose modified equation is degenerate; convergence is the content of Theorem 8.7. The agreement of constants is empirical evidence for the modified equation, not a proof: since (42) presupposes controlled derivatives that grow under refinement inside a layer of width $O(\Delta x)$, the expansion is formal there, and the sign change of the discrepancy across CFL numbers shows the agreement is only approximate.

8.4.5 A consistent scheme that selects the wrong solution

Consider the flux $\widehat{f}_{j+1/2} = f(U_j)$ if $\frac{1}{2}(U_j + U_{j+1}) > 0$ and $f(U_{j+1})$ otherwise, which is consistent and conservative but not monotone. For the Riemann data $U_j = -1$ ($j \le 0$), $U_j = +1$ ($j > 0$) every numerical flux equals $f(\pm 1) = \frac{1}{2}$, so $U^{n+1} = U^n$ exactly and the scheme returns the stationary upward jump, the analogue for these data of u_A in Section 5 (a weak solution by (18) with speed 0, but not an entropy solution). Godunov's flux at the same interface is $f(0) = 0$, and the resulting scheme produces instead the fan $u = x/t$ of Theorem 7.1. At $t = 1$ the $L^1(-2, 2)$ distance to the entropy solution is 1.000 for the non-monotone scheme for all $N = 200, 400, 800$, since $\int |\operatorname{sgn} x - \operatorname{clip}(x, -1, 1)|\,dx = 1$, while for Godunov's scheme it is 6.1×10^{-2}, 3.6×10^{-2}, 2.1×10^{-2}. By the Lax–Wendroff theorem [10, Ch. 12], if the approximations of a consistent conservative scheme remain bounded and converge a.e. (or in L^1_{loc}), the limit is a weak solution, so consistency and conservation alone do not select the entropy solution. Monotonicity (Theorem 8.7) is a sufficient condition for convergence to it, though not a necessary one for every scheme.

9 Conclusion

The weak formulation fixes the speed of a shock through the Rankine–Hugoniot condition, but it does not fix which discontinuities are admissible: already for Riemann data there are infinitely many weak solutions (Section 5). The Oleinik condition, whose local form at a jump is the Lax condition $u_\ell > u_r$, selects exactly one of them (Sections 6–7). This is the solution selected by the viscous Burgers regularisation: the Cole–Hopf solutions converge to it (Theorem 8.3, Remark 8.4), and a viscous travelling wave connecting u_ℓ to u_r exists if and only if $u_\ell > u_r$ (Proposition 8.6).

Godunov's scheme selects the same solution. Its shock error is $0.50\,\Delta x$ on aligned grids, and the modified equation (42) gives an effective viscosity of roughly $0.18\,\Delta x$ that vanishes with the mesh (a heuristic consistency check, since the expansion is formal in the shock layer), while a consistent conservative but non-monotone flux converges to the non-entropy weak solution (Section 8.4.5). The numerics are consistent with Theorem 8.7 [4], which supplies the convergence statement. Taken together, the results show that conservation determines the speed of a shock, while entropy, whether imposed through the Oleinik condition, through vanishing viscosity, or through a monotone numerical scheme, determines which shocks occur.

Several questions remain outside the scope of this paper. We restricted to bounded data and to the strictly convex flux $u^2/2$, where the Oleinik bound and the Lax condition are available; for general fluxes, systems and several space dimensions one must work with Kružkov's entropy inequalities or Oleinik's condition E, and uniqueness and convergence are then much harder. The rarefaction experiments did not settle the asymptotic rate of Godunov's scheme, and the agreement of the effective viscosity with the modified-equation prediction is empirical only. The picture established here for Burgers' equation, that discontinuities are admissible exactly when they are limits of viscous layers, is the model for the entropy conditions of more general conservation laws.

References

[1] H. Bateman. Some recent researches on the motion of fluids. *Monthly Weather Review*, 43(4):163–170, 1915.

[2] J. M. Burgers. A mathematical model illustrating the theory of turbulence. *Advances in Applied Mechanics*, 1:171–199, 1948.

[3] Julian D. Cole. On a quasi-linear parabolic equation occurring in aerodynamics. *Quarterly of Applied Mathematics*, 9(3):225–236, 1951.

[4] M. G. Crandall and A. Majda. Monotone difference approximations for scalar conservation laws. *Mathematics of Computation*, 34(149):1–21, 1980.

[5] Lawrence C. Evans. *Partial Differential Equations*. American Mathematical Society, 2nd edition, 2010.

[6] S. K. Godunov. A difference method for numerical calculation of discontinuous solutions of the equations of hydrodynamics. *Matematicheskii Sbornik*, 47(89)(3):271–306, 1959.

[7] Eberhard Hopf. The partial differential equation $u_t + uu_x = \mu u_{xx}$. *Communications on Pure and Applied Mathematics*, 3(3):201–230, 1950.

[8] S. N. Kružkov. First order quasilinear equations in several independent variables. *Mathematics of the USSR-Sbornik*, 10(2):217–243, 1970.

[9] Peter D. Lax. Hyperbolic systems of conservation laws ii. *Communications on Pure and Applied Mathematics*, 10(4):537–566, 1957.

[10] Randall J. LeVeque. *Finite Volume Methods for Hyperbolic Problems*. Cambridge University Press, 2002.

[11] M. J. Lighthill and G. B. Whitham. On kinematic waves ii: A theory of traffic flow on long crowded roads. *Proceedings of the Royal Society A*, 229(1178):317–345, 1955.

[12] Russell Ng. Source code for the numerical experiments (`godunov.py`, `make_godunov_figures.py`, `numerical_viscosity.py`, `cole_hopf_riemann.py`), 2026. GitHub repository, `https://github.com/luna-ortus-cor/MA4221-Project`.

[13] O. A. Oleinik. Discontinuous solutions of non-linear differential equations. *Uspekhi Matematicheskikh Nauk*, 12(3(75)):3–73, 1957. English translation: Amer. Math. Soc. Transl. Ser. 2, 26 (1963), 95–172.